Question ID: 63d03c0b

Question

For a particular car, the linear function f gives the predicted power, in brake horsepower (bhp), for engine speeds between $1{,}000 \text{ revolutions per minute } (\text{rpm})$ and $6{,}000 \text{ rpm}$. According to this function, the car's predicted power is 433 bhp at an engine speed of $3{,}331 \text{ rpm}$ and 600 bhp at an engine speed of $4{,}500 \text{ rpm}$. The equation $f(x) = \frac{1}{7}(x - a) + 433$ defines f, where x is the engine speed, in rpm, and a is a constant. What is the value of a?